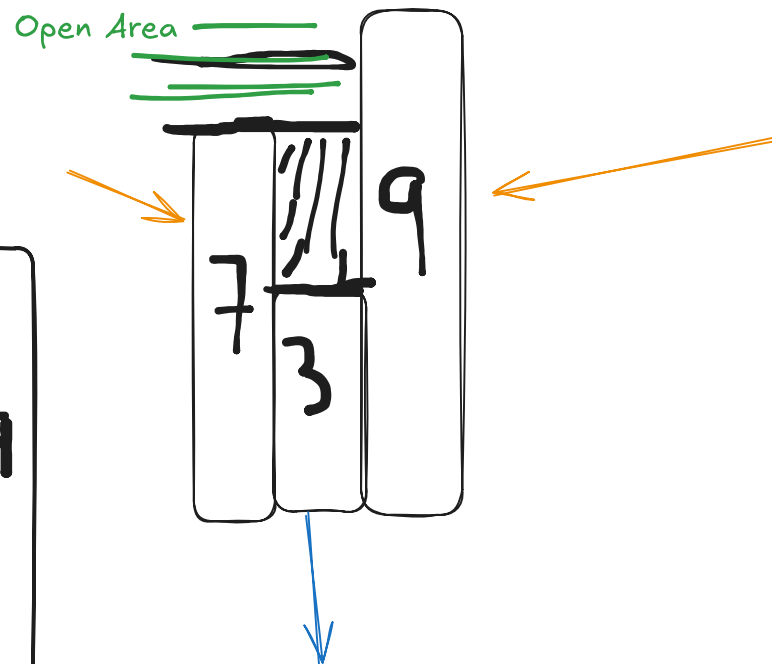
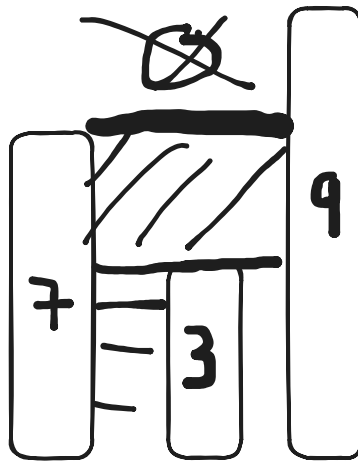


Topic: CodeKata-02
Date: 16 August 2025
Mentor: Sushant Goyal

You are given an array of non-negative integers representing height of walls at index i as A_i and the width of each block is 1.
Compute how much air can be encapsulated between the walls of chamber.

$A = [7, 3, 9]$

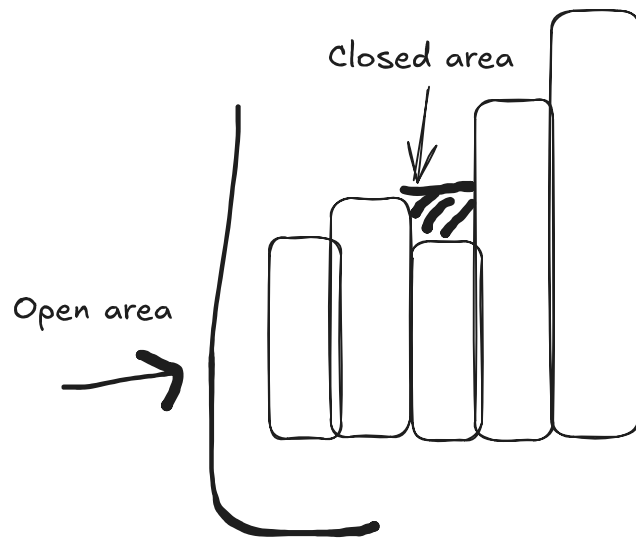
$A[i] \rightarrow$ height of the walls
width = 1



1. check for the smallest element
2. difference available between the smallest and second largest element - min of left and the right walls

air trapped = second largest element - smallest element
= $7 - 3$
= 4

$A = [3, 4, 3, 6, 7]$



Keep a track of the left and right elements to each of the element present in the array

left [] and right []

left[0] = A[0]

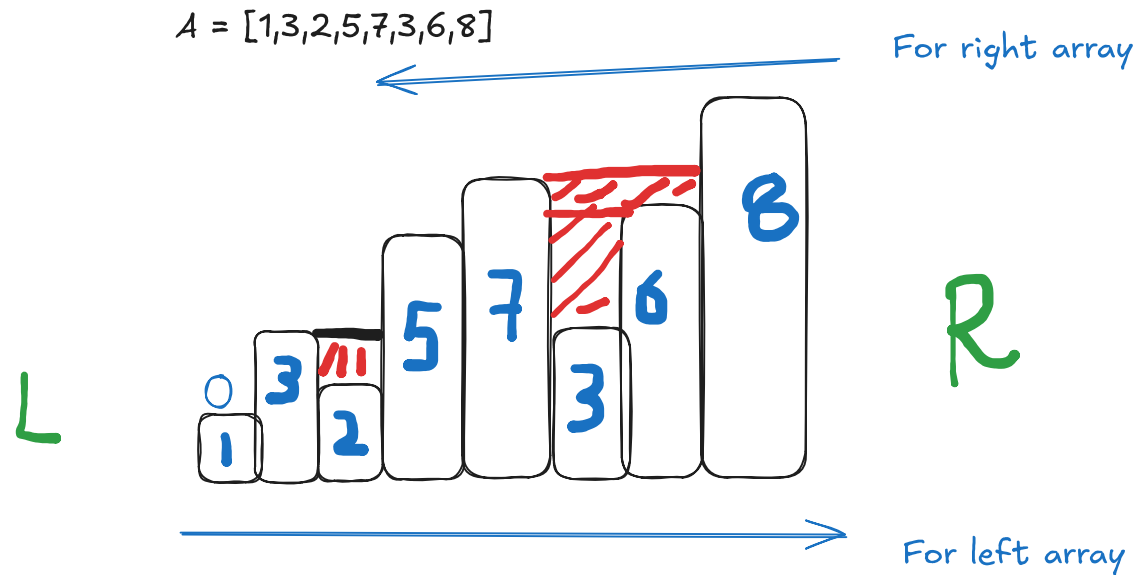
right[n-1] = A[n-1]

left[i] = maximum (previous element in the left array, element at current position in array)

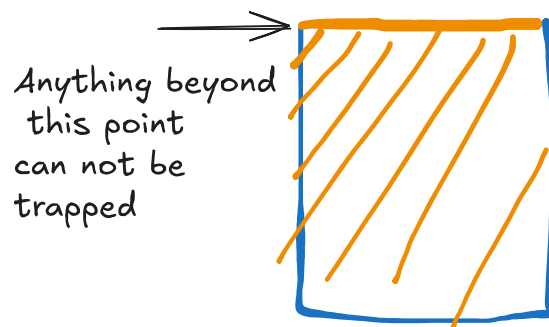
right[i] = maximum (next element on right, element at current position in array)

left and right array in place. we will calculate the air trapped

sum of the differences between the left and right elements from the current element



$left[0] = 1$ // $[1, 0, 0, 0, 0, 0, 0, 0]$
 $right[n-1] = 8$ // $[0, 0, 0, 0, 0, 0, 0, 8]$
 (max of $left[i-1]$ and height array)
 $left[1] = 3$ (max of 1 and 3)
 $left[2] = 3$ (max of 3 and 2)
 $left[3] = 5$ (max of 3 and 5)
 $left[4] = 7$ (max of 5 and 7)
 $left[5] = 7$ (max of 7 and 3)
 $left[6] = 7$ (max of 7 and 6)
 $left[7] = 8$ (max of 7 and 8)



First Occurrence:
 smallest height = 2
 second highest = 3
 difference = $3 - 2 = 1$

Second Occurrence:
 smallest height = 3
 second highest = 6
 difference = $6 - 3 = 3$

Third Occurrence:
 smallest height = 6
 second highest = 7
 difference = $7 - 6 = 1$

Fourth Occurrence:

smallest = 6 (3 = height of wall, 3 from difference of second occurrence)

second highest = 7

difference = $7 - 6 = 1$

Total AIR Trapped = $1 + 3 + 1 + 1 = 6$

$A = [1, 3, 2, 5, 7, 3, 6, 8]$

$\text{right}[n-1] = 8$

Max of right element $\text{right}[i+1]$ and current height element

$\text{right}[n-2] = 8$ (max of 8 and 6)

$\text{right}[n-3] = 8$ (max of 8 and 3)

$\text{right}[n-4] = 8$ (max of 8 and 7)

$\text{right}[n-5] = 8$ (max of 8 and 5)

$\text{right}[n-6] = 8$ (max of 8 and 2)

$\text{right}[n-7] = 8$ (max of 8 and 3)

$\text{right}[n-8] = 8$ (max of 8 and 1)

$i=0$ $\text{left}[0] = 1$, $\text{right}[0] = 8$

Minimum of left and right elements - height of wall at 0 position ($\text{arr}[0]$)

$i=0$ $\text{Min}(1, 8) - 1 = 0$

$i=1$ $\text{Min}(3, 8) - 3 = 0$

$i=2$ $\text{Min}(3, 8) - 2 = 1$

$i=3$ $\text{Min}(5, 8) - 5 = 0$

$i=4$ $\text{Min}(7, 8) - 7 = 0$

$i=5$ $\text{Min}(7, 8) - 3 = 4$

$i=6$ $\text{Min}(7, 8) - 6 = 1$

$$i=7 \text{ Min}(8,8) - 8 = 0$$

airTrapped = sum of result at each iteration